Project Documentation: AlgorySoft E-Commerce Platform

Project Name: AlgorySoft Pre-built Project Marketplace

Version: 1.0

Prepared For: Development Team & Stakeholders_____

1. Project Overview

The AlgorySoft platform is a specialized e-commerce solution designed to sell pre-built software projects (React, Node.js, etc.) along with tiered customization services. The system facilitates the sale of "Products" (code templates) and "Services" (customization) through a streamlined workflow involving client interactions, payment gateways, and an administrative project management suite.

Key Features

- Client Side: Product browsing, tiered purchasing (Basic/Silver/Gold), custom request handling, real-time order tracking, and live chat.
- Admin Side: Product management (CRUD), order lifecycle management, request approval workflows, 1:1 client communication, and analytics.

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2. Theoretical Data Models

Before diving into the flow, the following data structures define how information is stored and processed.

A. User Schema (Client)

- id: Unique Identifier
- name, email, password_hash, phone
- role: "Client" (Default)
- wishlist: Array of Product IDs

B. Product Schema

- id: Unique Identifier
- title, description, images[], tech_stack[] (e.g., React, MongoDB)
- live_preview_url: Link to demo
- pricing_tiers:

o Basic: { price: 5000, features: ["Static", "Contact Form", "Domain"] }

o Silver: { price: 12000, features: ["Dynamic", "Admin Panel", "Auth"] }

o Gold: { price: 25000, features: ["AWS Deploy", "Payment Gateway", "Full Support"] }

C. Order Schema

- id: Unique Order ID
- user_id: Reference to Client
- product_id: Reference to Product
- selected_tier: "Basic" | "Silver" | "Gold"
- status: "Pending" | "Processing" | "Deploying" | "Delivered"
- progress_percentage: 0-100
- payment_status: "Paid" | "Requested"
- discount_applied: Boolean (2% if immediate pay)

D. Request Schema (Custom Orders)

- id: Unique Request ID
- user_id: Reference to Client
- requirements_text: String
- attachments[]: File URLs
- status: "New" | "Approved" | "Quoted" | "In Development"
- admin_notes: String

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3. Client Side Workflow & Data Flow

3.1 Navigation & Discovery

Pages: Home, Shop, Product Details, Cart, Checkout.

1. Landing Page: Displays Trending Products, Hot Products, and Carousel.
2. Shop Page: Grid view of products with Filters (Category, Tech Stack, Price).
3. Authentication: Login/Register required for "Wishlist", "My Orders", and "Checkout".

3.2 Product Detail & Purchase Flow

Logic Flow:

1. User clicks a Product Card

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Navigates to Product Detail Page.

2. Detail Page Displays:

- Images, Description, Tech Stack, Features.
- Live Preview Button: Opens iframe/modal.
- Pricing Tiers: Cards showing Basic, Silver, Gold prices.

3. Purchase Decision:

- Path A (Instant Purchase):
- User selects a tier (e.g., Silver).
- Clicks "Buy Now".
- Logic Check: If paying immediately

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Apply 2% Discount.

- Payment Gateway opens.

o Success

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Order Created in DB with status "Processing"

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Redirect to "My Orders".

o Path B (Request Customization):

o User clicks "Request Custom".

o Logic Check: Is User Logged In?

o No: Redirect to Login, then back to form.

o Yes: Open Requirement Form.

o User fills details, uploads files.

o Clicks Submit

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Request Entry Created in DB

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Email sent to Admin.

3.3 Post-Purchase: Order Tracking

Pages: My Orders, Order Detail.

- Order List: Shows all purchased projects.
- Order Detail:

o Progress Bar (e.g., "Design: 100%", "Dev: 45%").

o Current Status (Processing / Deploying).

o Chat Interface: 1-to-1 messaging with Admin for project updates.

o File Sharing: Ability to share requirement docs/images within chat.

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4. Admin Side Workflow & Data Flow

4.1 Dashboard Overview

- Stats: Revenue Today, Active Orders, Pending Requests.
- Quick Actions: Approve Requests, Reply to Chats.

4.2 Product Management (CRUD)

- Create Product: Input Name, Description, Tech Stack, Upload Images.
- Define Tiers: Set specific prices and feature lists for Basic/Silver/Gold cards for that specific product.
- Search & Pagination: Server-side search to handle large inventories.

4.3 Request & Approval Pipeline

Flow:

1. New Request Notification: Admin sees new "Custom Request" in the Request Tab.
2. Review: Admin reads requirements and checks user history.
3. Action:

o Approve: Admin clicks "Approve"

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Status changes to "Quoted" or "Processing".

o Talk: Admin initiates Chat from the Request Detail page to clarify scope.

4. Conversion: Once approved, the Request can be converted into an active Project/Order for tracking.

4.4 Order Management & Progress Tracking

Flow:

1. Admin selects an Order ID.

2. Update Progress: Slider/Input to update percentage (e.g., 10% to 50%).

3. Update Status: Dropdown (e.g., "In Progress"

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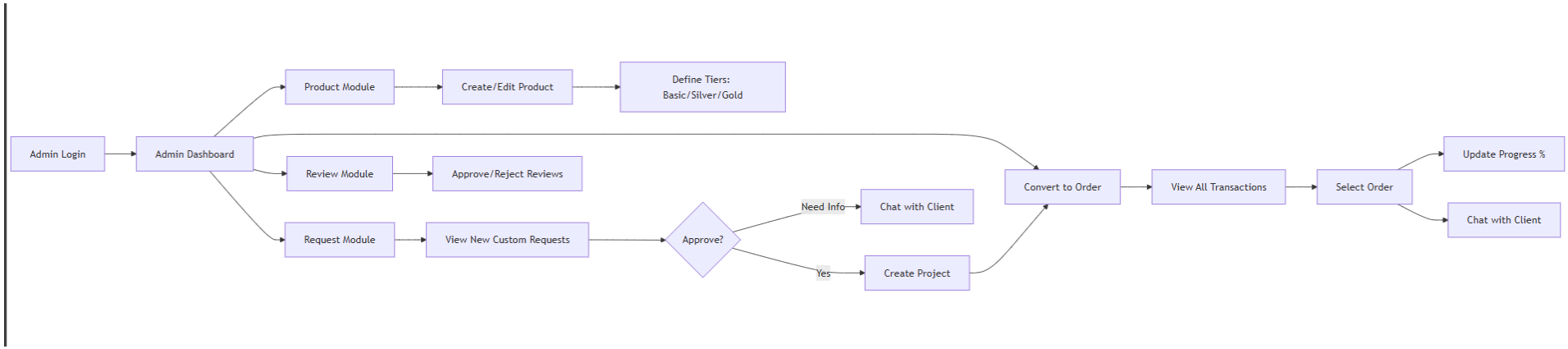
"Testing").

4. Client Notification: System automatically notifies the client via email/panel when progress updates.

5. Chat: Admin can send messages/files to the client regarding the specific order.

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5. Visual Data Flow Diagrams (DFD)


Admin Login
Admin Dashboard
Product Module
Create/Edit Product
Define Tiers:
Basic/Silver/Gold
Review Module
Approve/Reject Reviews
Request Module
View New Custom Requests
Approve?
Need Info
Chat with Client
Yes
Create Project
Convert to Order
View All Transactions
Select Order
Update Progress %
Chat with Client

Client Lands on Home
Clicks Product
Product Detail Page
Decision Point
Need Customization
Standard Part
Click Request Button
Logged In?
Yes
No
Login/Register
Submit Requirement Form
Email Notification to Admin
Admin Approval
Add to Cart / Buy Now
Payment Type
Request Quote
Pay Immediately
Payment Later
Apply 2% Discount
Payment Gateway
Success
Order Created in DB
My Orders Page
Track Progress & Chat

A. Tiered Pricing Logic (The Cards)

- Basic Card (₹5k Range):
  - Static Frontend (HTML/CSS/JS or basic React).
  - Contact Form integration.
  - Basic Domain assistance.
  - No Database/Admin Panel.
- Silver Card (₹10k+ Range):
  - Dynamic Frontend.
  - Custom Admin Dashboard.
  - Database Integration (MongoDB/Firebase).
  - Authentication System.
- Gold Card (₹20k+ Range):
  - Full Stack Application.
  - AWS/Cloud Deployment.
  - Payment Gateway Integration.
  - Priority Support.

B. Communication Integration

- WhatsApp: Floating button on Client Side for quick support.
- Gmail/LinkedIn: Footer links.
- Internal Chat: Socket.io or similar technology for real-time 1-to-1 chat between Admin and Client inside the Order Tracking page.

C. Automation Rules

1. Discount Logic: Total Price * 0.98 applied at checkout if payment method is "Instant Pay".

2. Status Update: When Admin sets progress to 100%, system automatically asks user for a Review.

3. Login Guard: Any action involving PII (Personally Identifiable Information) or financial transaction triggers the Login Guard.

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7. Technical Requirements Summary

- Frontend: React.js (Client & Admin Panel), CSS/Tailwind.
- Backend: Node.js with Express.
- Database: MongoDB (for flexible product schemas) or PostgreSQL.
- Real-time: Socket.io (Chat & Progress Updates).
- Storage: AWS S3 or Cloudinary (for product images and requirement files).
- Payments: Razorpay or Stripe API.